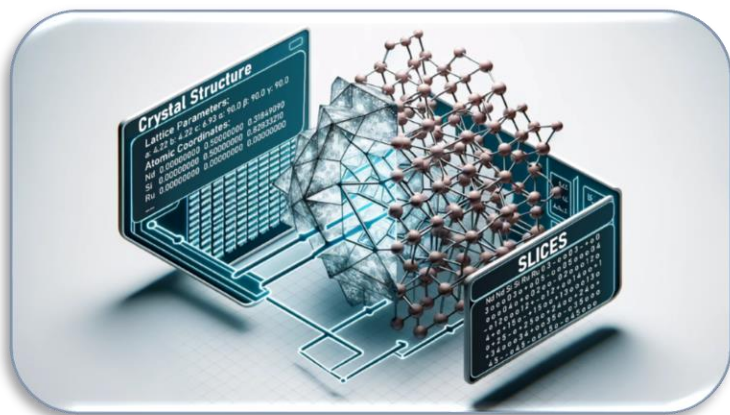




基于MatterGPT的多目标新材料逆向设计



SLICES晶体语言



大语言模型

肖 航 嶺南大學 跨学科学院

• 生成式AI：AI的iPhone时刻

背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT



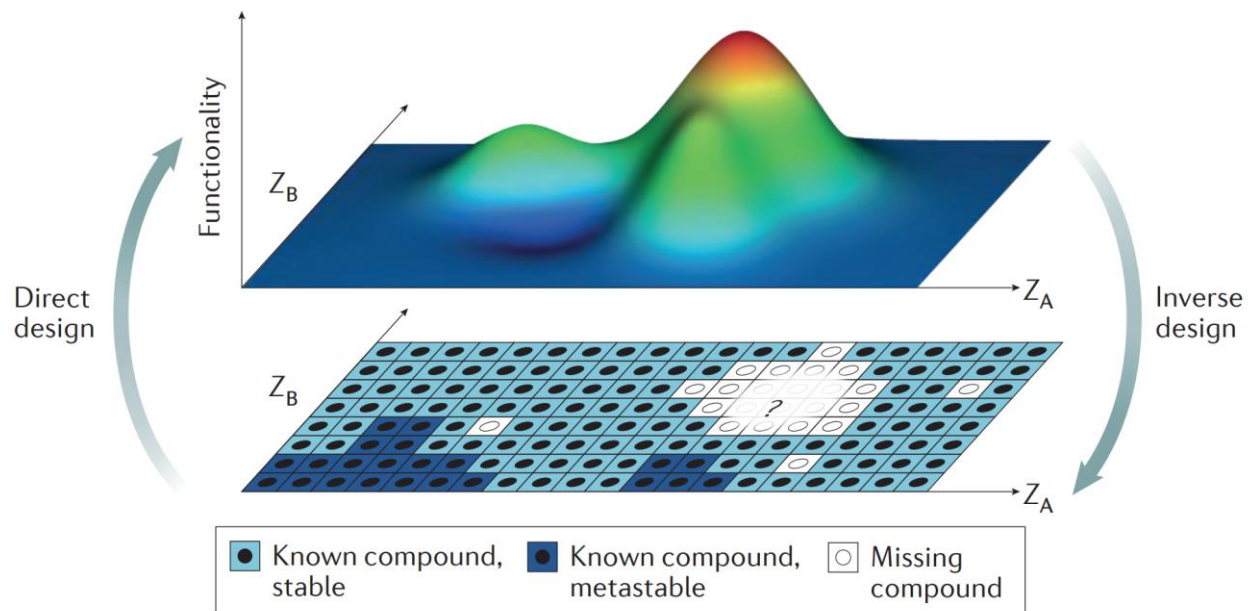
• 逆向设计与正向设计

背景介绍



SLICES
晶体编码

SLICES
晶体解码



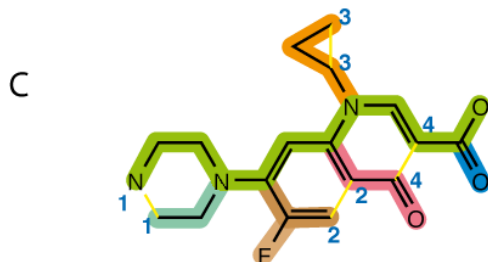
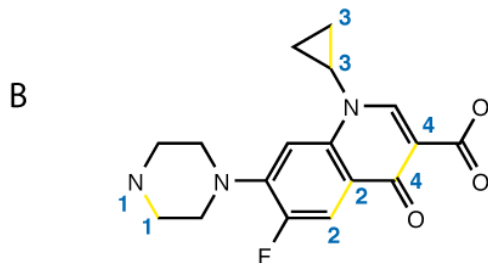
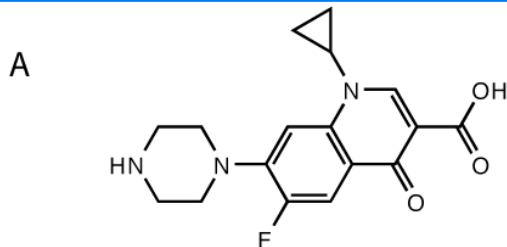
MatterGPT

Zunger, Alex. "Inverse design in search of materials with target functionalities." *Nature Reviews Chemistry* 2.4 (2018): 0121.

• 痛点：可逆晶体表示法

背景介绍

可逆分子表示法 SMILES



D

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

■ 可逆性:

分子结构 $\rightleftharpoons$ SMILES字符串

■ 对称不变性:

平移、旋转、轮换不变性

■ 分子：准3D；非周期性；元素集中

■ 晶体：3D；周期性；元素分散

可逆晶体表示法？

SLICES
晶体编码

SLICES
晶体解码

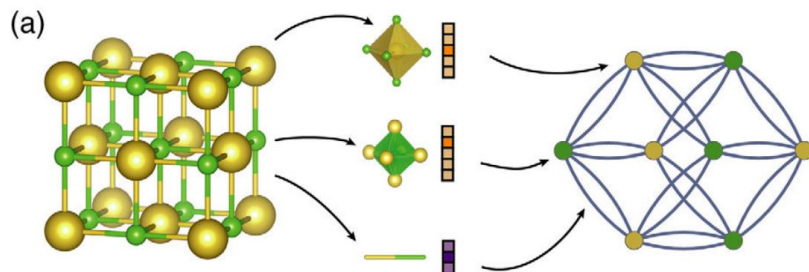
MatterGPT

• 前人工作

背景介绍

晶体图 谢天和Grossman 2018

晶体图卷积神经网络(CGCNN)

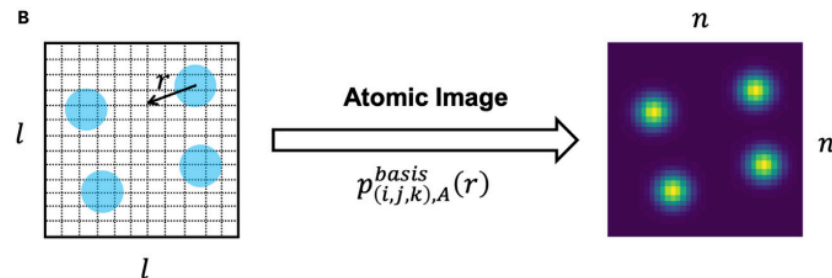


满足对称不变性

不可逆

3D图像 Noh团队 2019

通过投影获得图像



不满足对称不变性

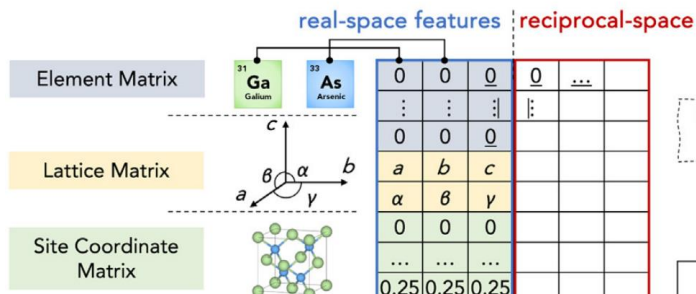
计算成本高

SLICES
晶体编码

SLICES
晶体解码

FTCP Zekun Ren 团队 2022

直接使用晶格常数和原子坐标等信息

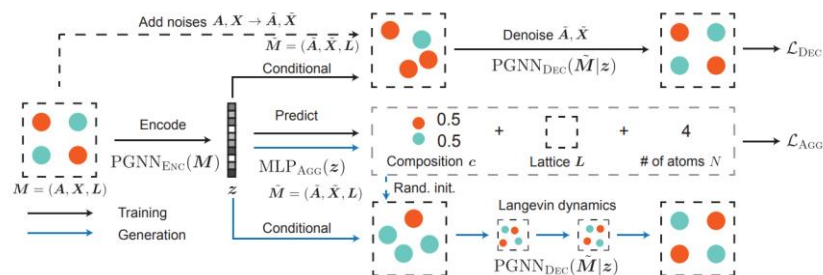


不满足对称不变性

可逆性较低 69.9%

CDVAE 谢天团队 2022

基于晶体图和扩散模型, 目标生成稳定材料



可逆性低 45.4%

MatterGPT

来自SMILES的启发

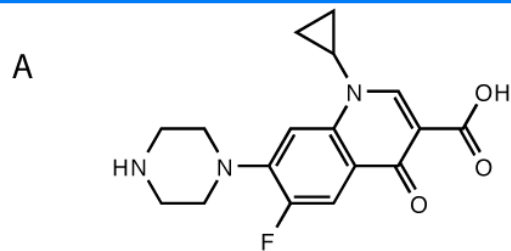
背景介绍

SLICES
晶体编码

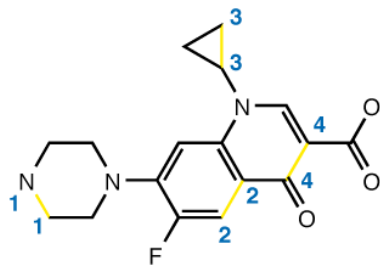
SLICES
晶体解码

MatterGPT

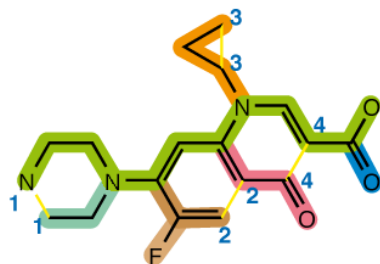
可逆分子表示法 SMILES



分子图



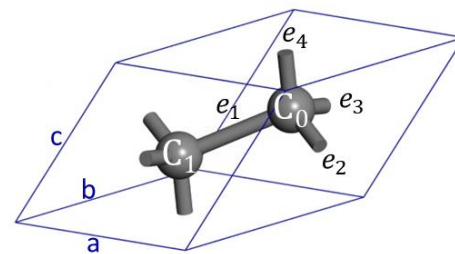
C



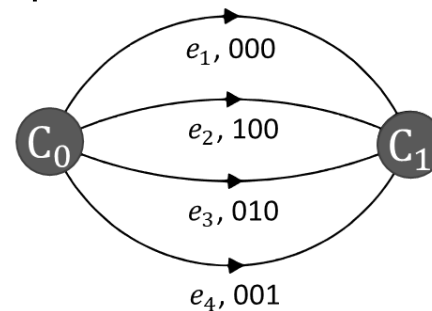
D

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

可逆晶体表示法 SLICES



标记商图

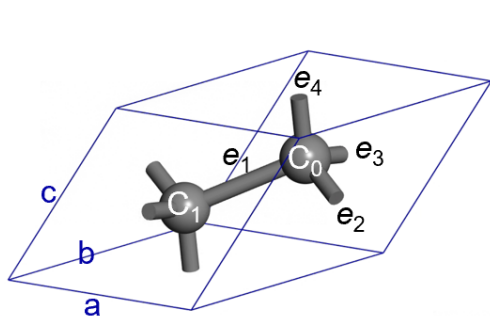


e_1	e_2	e_3	e_4
C C 0 1 000 0 1 +00 0 1 o+o 0 1 oo+			
元素符号	原子序号	化学键偏移向量	

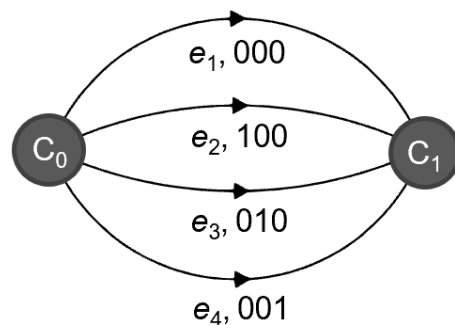
• SLICES字符串的构成

背景介绍

■ 化学键识别算法: EconNN 或 CrystalNN



晶体



晶体图

Diagram illustrating the SLICES string structure, showing the mapping from crystal structure to graph representation and then to the string format.

e ₁		e ₂		e ₃		e ₄													
C	C	0	1	0	0	0	1	+	0	0	1	0	+	0	1	0	0	+	
元素符号		原子序号		化学键偏移向量															

SLICES 字符串

SLICES
晶体解码



元素符号: 晶胞内的原子

节点编号: 一个原子对应一个编号, 晶胞内有几个原子, 编号就有几个

化学键偏移矢量: 用 o、+、- 来表示偏移矢量中的 0、1、-1

MatterGPT

• SLICES晶体表示法与其他文本晶体表示法的对比

背景介绍

SLICES 晶体编码

SLICES 晶体解码

MatterGPT

representation name	representation
Composition	MgPtTa
Atom Sequence	Mg Ta Pt
Atom Sequence ++	Mg Ta Pt 4.32 4.32 4.32 60 60 60
SLICES	Mg Ta Pt 0 2 - - o 0 2 - o - 0 2 - o o 0 2 o - - 0 2 o - o 0 2 o o - 0 1 - - o 0 1 - o - 0 1 o - - 1 2 o o o
Crystal-text-LLM	3.5 4.2 4.4
	90 90 90
	Ta
	0.76 0.12 0.00
	Ta
	0.00 0.12 0.18
CIF P_1	V
	0.00 0.00 0.00
	Ga
	0.76 0.00 0.18
	data_MgTaPt
	_symmetry_space_group_name_H-M 'P 1'
CIF P_1	_cell_length_a 4.32
	_cell_length_b 4.32
	_cell_length_c 4.32
	_cell_angle_alpha 60.0
	_cell_angle_beta 60.0
	_cell_angle_gamma 60.0
CIF P_1	_symmetry_Int_Tables_number 1
	_chemical_formula_structural MgTaPt
	_chemical_formula_sum 'Mg1 Ta1 Pt1'
	_cell_volume 56.85
	_cell_formula_units_Z 1
	loop_
CIF P_1	_symmetry_equiv_pos_site_id
	_symmetry_equiv_pos_as_xyz
	1 'x, y, z'
	loop_
	_atom_site_type_symbol
	_atom_site_label
CIF P_1	_atom_site_symmetry_multiplicity
	_atom_site_fract_x
	_atom_site_fract_y
	_atom_site_fract_z
	_atom_site_occupancy
	Mg Mg0 1 0.0 0.0 0.0 1.0
Z-Matrix	Ta Ta2 1 0.58 0.58 0.58 1.0
	Pt Pt1 1 0.53 0.53 0.53 1.0
	Mg
	Ta 1 6.1
	Pt 2 0.5 1 0

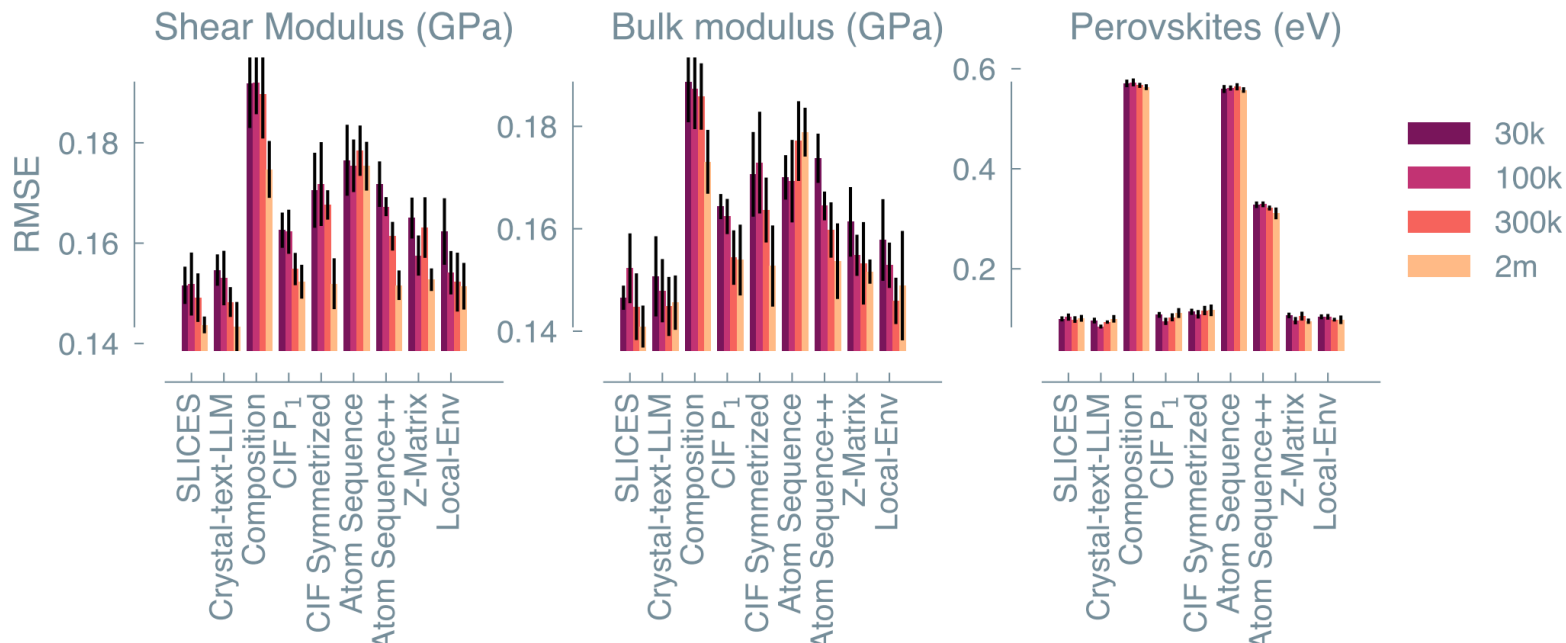
CIF-Symmetrized	data_MgTaPt
	_symmetry_space_group_name_H-M R3m
	_cell_length_a 4.32
	_cell_length_b 4.32
	_cell_length_c 10.57
	_cell_angle_alpha 90.0
CIF-Symmetrized	_cell_angle_beta 90.0
	_cell_angle_gamma 120.0
	_symmetry_Int_Tables_number 160
	_chemical_formula_structural MgTaPt
	_chemical_formula_sum 'Mg3 Ta3 Pt3'
	_cell_volume 170.55
CIF-Symmetrized	_cell_formula_units_Z 3
	loop_
	_symmetry_equiv_pos_site_id
	_symmetry_equiv_pos_as_xyz
	1 'x, y, z'
	2 '-y, x-y, z'
CIF-Symmetrized	3 '-x+y, -x, z'
	4 '-y, -x, z'
	5 '-x+y, y, z'
	6 'x, x-y, z'
	7 'x+1/3, y+2/3, z+2/3'
	8 '-y+1/3, x-y+2/3, z+2/3'
CIF-Symmetrized	9 '-x+y+1/3, -x+2/3, z+2/3'
	10 '-y+1/3, -x+2/3, z+2/3'
	11 '-x+y+1/3, y+2/3, z+2/3'
	12 'x+1/3, x-y+2/3, z+2/3'
	13 'x+2/3, y+1/3, z+1/3'
	14 '-y+2/3, x-y+1/3, z+1/3'
CIF-Symmetrized	15 '-x+y+2/3, -x+1/3, z+1/3'
	16 '-y+2/3, -x+1/3, z+1/3'
	17 '-x+y+2/3, y+1/3, z+1/3'
	18 'x+2/3, x-y+1/3, z+1/3'
	loop_
	_atom_site_type_symbol
CIF-Symmetrized	_atom_site_label
	_atom_site_symmetry_multiplicity
	_atom_site_fract_x
	_atom_site_fract_y
	_atom_site_fract_z
	_atom_site_occupancy
CIF-Symmetrized	Mg Mg0 3 0.0 0.0 0.0 1.0
	Ta Ta1 3 0.0 0.0 0.42 1.0
	Pt Pt2 3 0.0 0.0 0.47 1.0
	R3m
Local Env	Ta (1a) [Ta]#[Pt]
	Pt (1a) [Ta]#[Pt]
	Mg (1a) [Ta] [Mg] [Ta] . [Ta] . [Pt] . [Pt]
Robocrys	MgTaPt crystallizes in the trigonal R3m space group.
	Mg(1) is bonded in a 6-coordinate geometry to three
	equivalent Ta(1) and three equivalent Pt(1) atoms. All
	Mg(1)-Ta(1) bond lengths are 2.66 Å. All Mg(1)-Pt(1) bond
	lengths are 3.22 Å. Ta(1) is bonded in a single-bond
	geometry to three equivalent Mg(1) and one Pt(1) atom.
Robocrys	The Ta(1)-Pt(1) bond length is 0.55 Å. Pt(1) is bonded in
	a single-bond geometry to three equivalent Mg(1) and one
	Ta(1) atom.

Alampara, N.; Miret, S.; Jablonka, K. M. MatText: Do Language Models Need More than Text & Scale for Materials Modeling? arXiv June 28, 2024. <http://arxiv.org/abs/2406.17295> (accessed 2024-08-19).

• SLICES晶体表示法与其他文本晶体表示法的对比

背景介绍

■ SLICES表示法在性质预测上的优异表现



SLICES
晶体编码

SLICES
晶体解码

is even more striking. Conventionally, one might expect that adding more information improves predictive performance. For instance, one would expect that the addition of positional information increases performance in predicting the bulk and shear modulus (which one also observes for other models, e.g., GNNs, on these tasks). In our experiments, however, we find that this is not the case. **In several instances, we could achieve comparable, if not better, performance with less information using a representation such as SLICES, which only contains information about the composition and the bond network.** Notably, the predictive performance also does not improve when we incorporate classical

MatterGPT

• 实现可逆：重建晶体结构

背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

- **一. 基于Eon的图论方法生成初始猜测结构**
纯数学的初始猜测
- **二. 基于定制化的GFN-FF估计键长键角**
化学信息修饰
- **三. 深度学习力场M3GNet IAP进行结构优化**
进一步提高还原度

• 实现可逆：重建晶体结构（一）

背景介绍

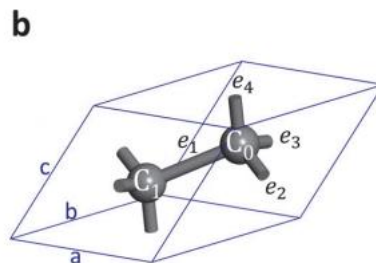
SLICES
晶体编码

SLICES
晶体解码

MatterGPT

■ 一. 基于Eon的图论方法生成初始猜测结构

纯数学的初始猜测



$$\Omega = B^{-1} \cdot \alpha$$

$$B_{dia} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \dots \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$



$$\begin{pmatrix} \cdot b & a \cdot c \\ \cdot b & b \cdot c \\ \cdot b & c \cdot c \end{pmatrix}, \Omega(L^*) = \begin{pmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \dots \\ \vec{e}_m \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

$$Z = L_A \cdot P \cdot (L_A)^t \quad Z_{dia} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix} \cdot I \cdot \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

• 实现可逆：重建晶体结构（一）

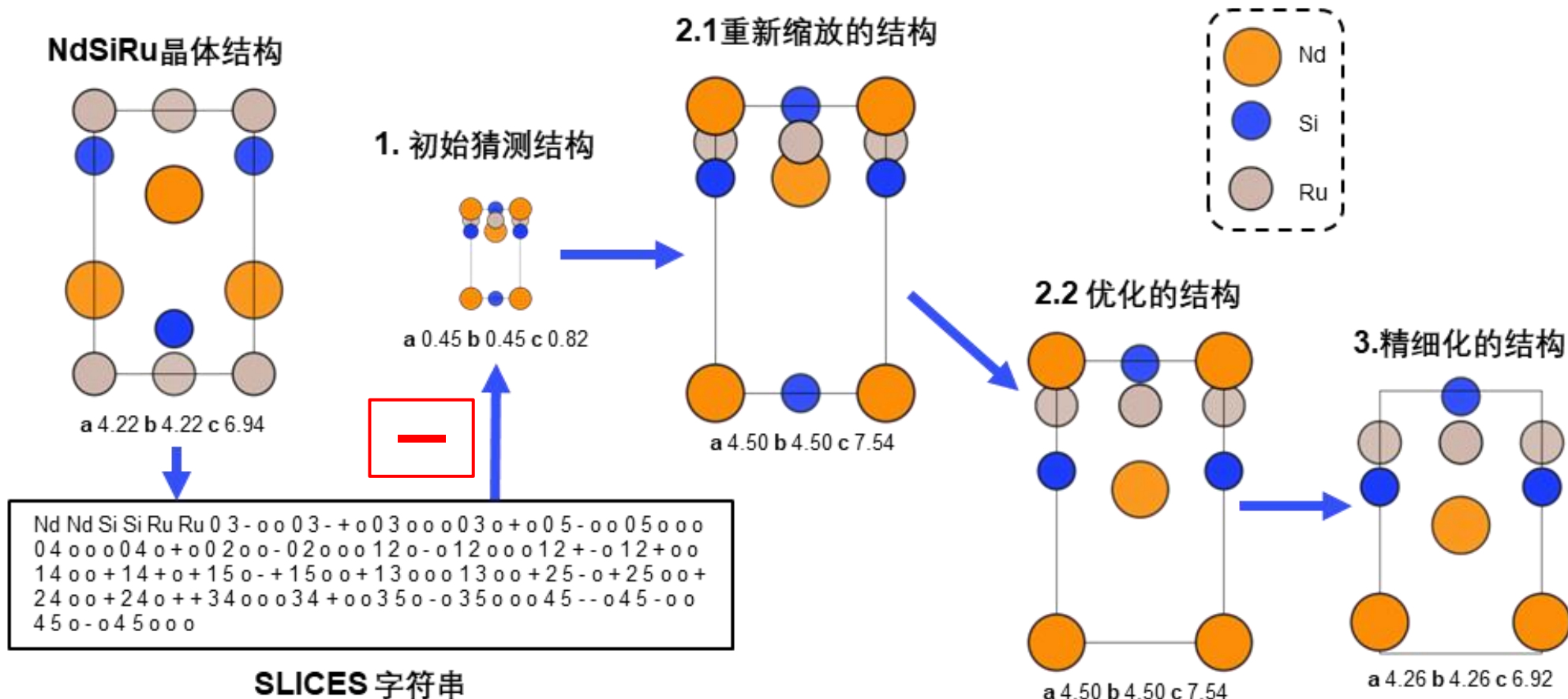
背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

■ 一. 基于Eon的图论方法生成初始猜测结构



• 实现可逆：重建晶体结构（二）

背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

■ 一. 基于Eon的图论方法生成初始猜测结构

纯数学的初始猜测

■ 二. 基于定制化的GFN-FF估计键长键角

化学信息修饰

■ 三. 深度学习力场M3GNet IAP进行结构优化

进一步提高还原度

• 实现可逆：重建晶体结构（二）

背景介绍

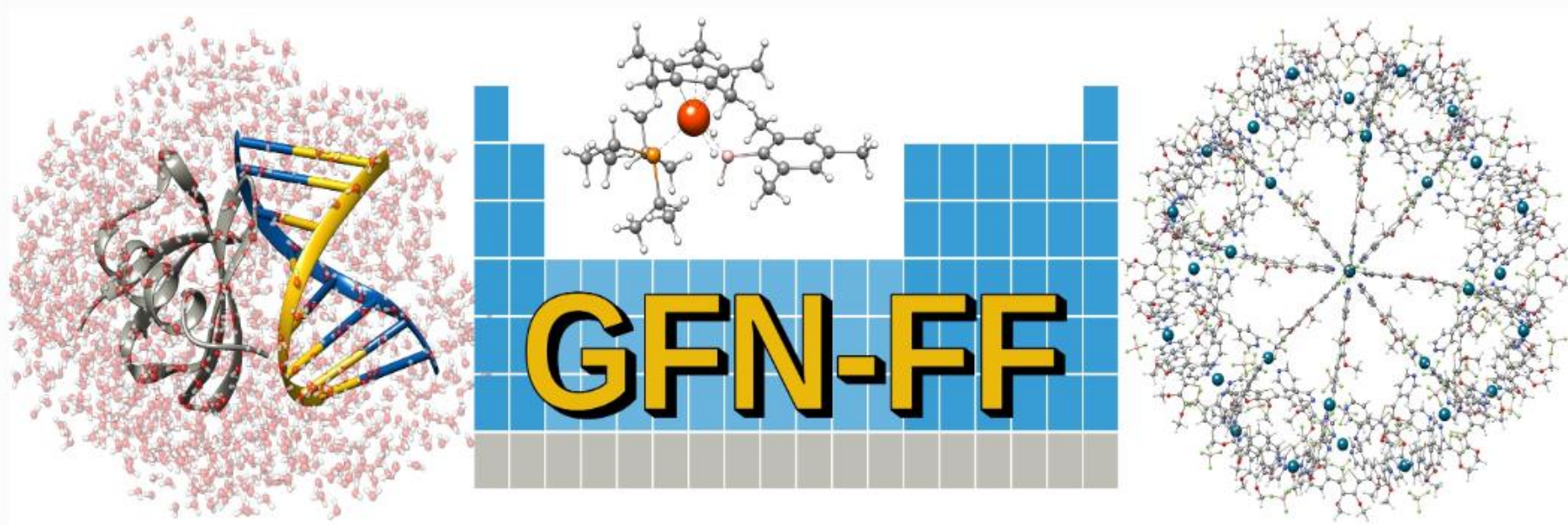
■ 二. 基于定制化的GFN-FF估计键长键角

基于拓扑结构和化学构成的键长键角预测

SLICES
晶体编码

SLICES
晶体解码

MatterGPT



Spicher, S. & Grimme, S. Robust atomistic modeling of materials, organometallic, and biochemical systems. *Angew. Chem. Int. Ed.* 59, 15665–15673 (2020)

• 实现可逆：重建晶体结构（二）

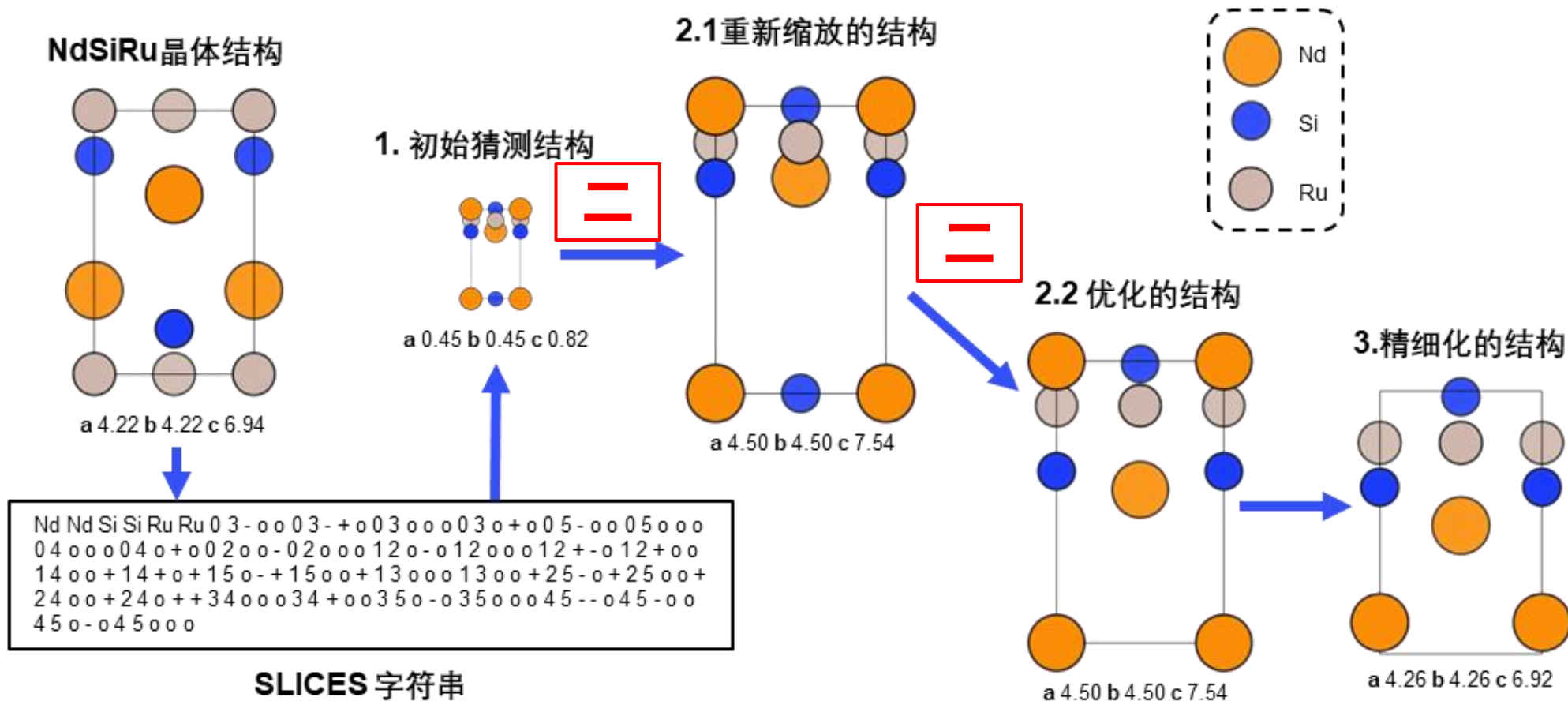
背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

■ 二. 基于定制化的GFN-FF估计键长键角



• 实现可逆：重建晶体结构（三）

背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

■ 一. 基于Eon的图论方法生成初始猜测结构

纯数学的初始猜测

■ 二. 基于定制化的GFN-FF估计键长键角

化学信息修饰

■ 三. 深度学习力场M3GNet IAP进行结构优化

进一步提高还原度

• 实现可逆：重建晶体结构（三）

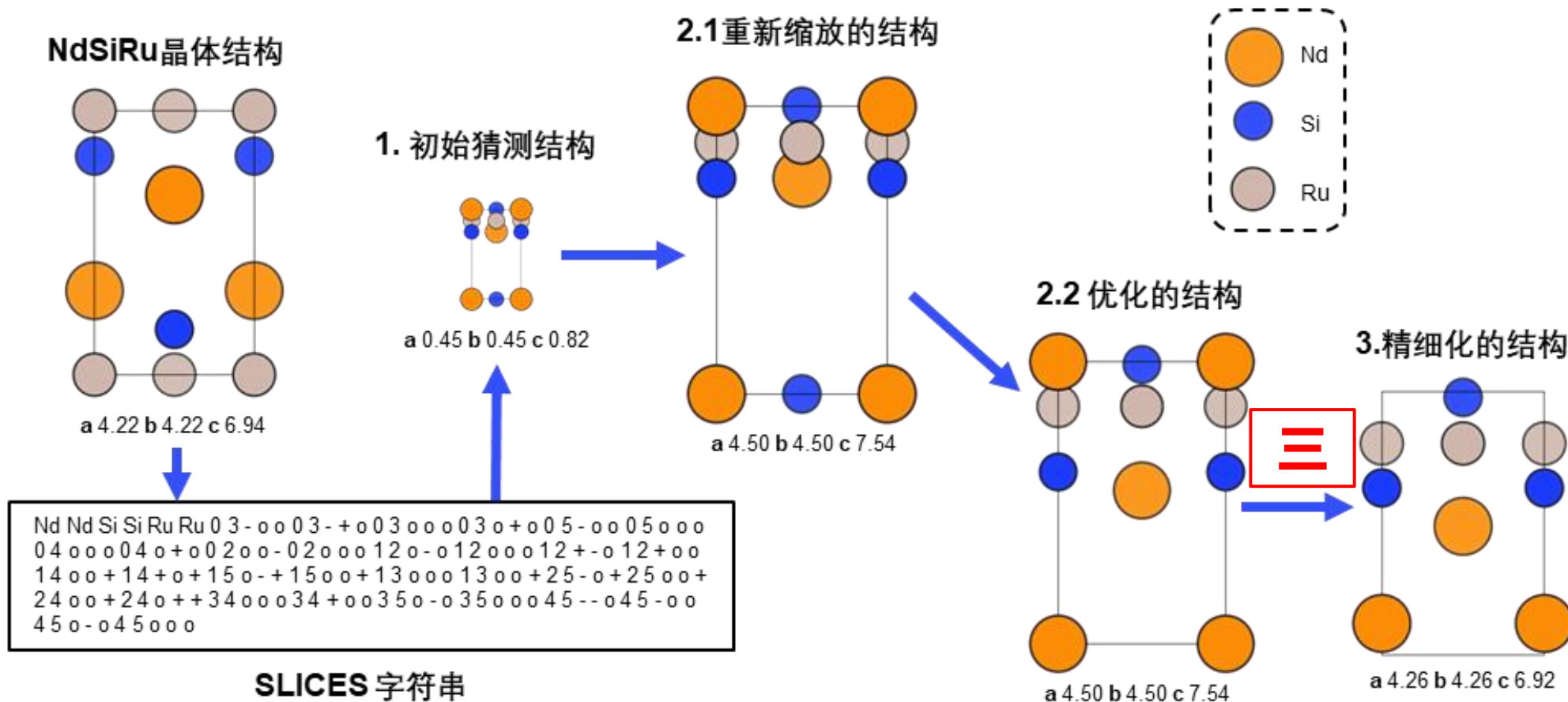
背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

■ 三. 深度学习力场M3GNet IAP进行结构优化



Chen, C. & Ong, S. P. A universal graph deep learning interatomic potential for the periodic table. Nat. Comput. Sci. 2,718–728 (2022).

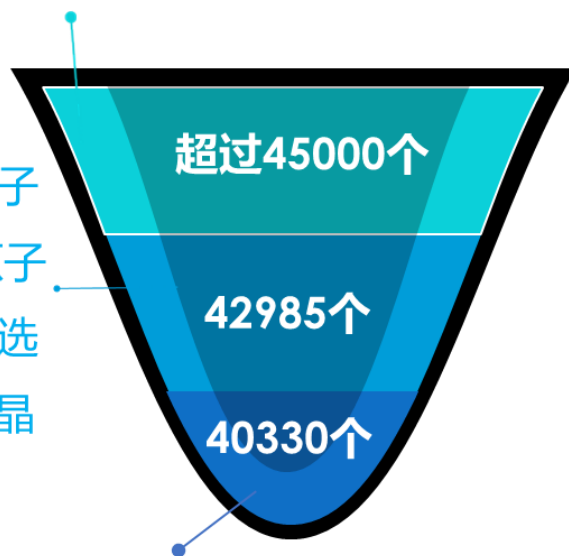
• 晶体重建性能

背景介绍

由Xie等人建立的MP-20数据集

超过45000个不同的晶体结构，
涵盖了元素周期表中的89种元素。

改进的GFN-FF，
不适用于含有原子
序数大于86的原子
的晶体结构。筛选
后还有42985个晶
体（95.04%）。



步骤(I)中使用的方法不能用于低维结构晶体，
筛选后还有40330个晶体(89.17%)。

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

筛选后的MP-20数据集的重建性能(M3GNet)

Setting	Match rate (%)		
	Rescaled structure	ZL^* - optimized structure	IAP-refined structure
Normal	91.36	94.05	94.95

筛选后的MP-20数据集的重建性能(CHGNet)

Setting	Match rate (%)		
	Rescaled structure	ZL^* - optimized structure	IAP-refined structure
Normal	91.36	94.05	95.97

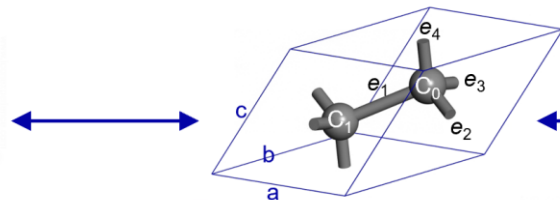
• 总结

背景介绍

SLICES:



Diamond



Atomic structure

```
C C 0 1 000 0 1 +00 0 1 0+0 0 1 00+
```

SLICES text

<https://github.com/xiaohang007/SLICES>



docker



SLICES
晶体解码

nature communications



Hugging Face

Article

<https://doi.org/10.1038/s41467-023-42870-7>

An invertible, invariant crystal representation for inverse design of solid-state materials using generative deep learning

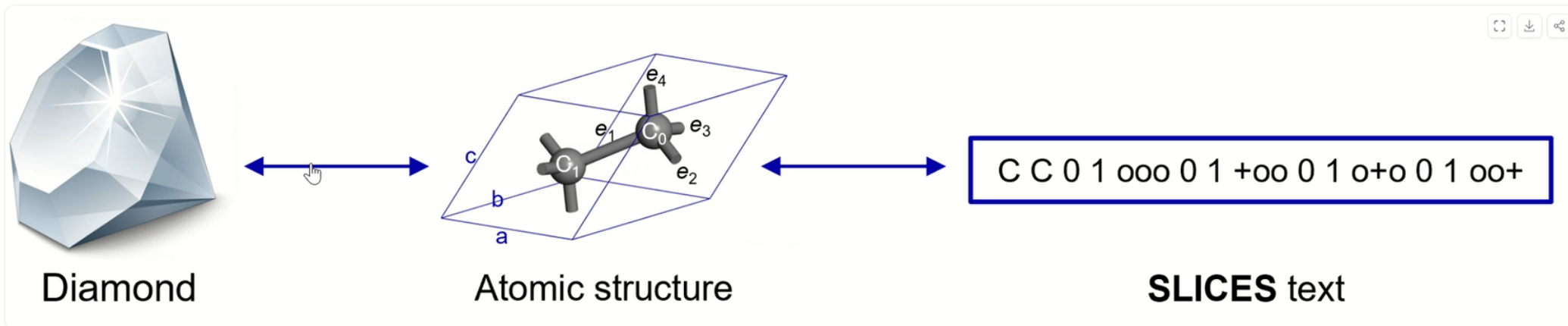
MatterGPT

• Huggingface Space 在线演示

Browser address bar: <https://huggingface.co/spaces/xiaohang07/SLICES>

Navigation bar: Spaces | xiaohang07 / SLICES | like 2 | Running | App | Files | Community | Settings

Crystal Structure and SLICES Converter



SLICES provides a text-based encoding of crystal structures, allowing for efficient manipulation and generation of new materials.

CIF-SLICES Conversion

SLICES Augmentation and Canonicalization

Choose CIF source

☒ Use example CIF (NdSiRu.cif)

☐ Upload custom CIF

Convert CIF to SLICES

SLICES String

Download CIF

Enter SLICES String

· 前人工作：新材料逆向设计方法

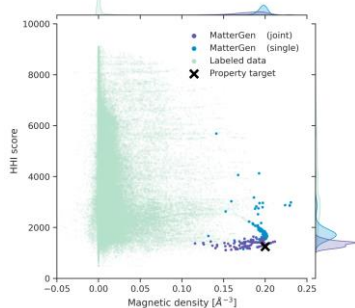
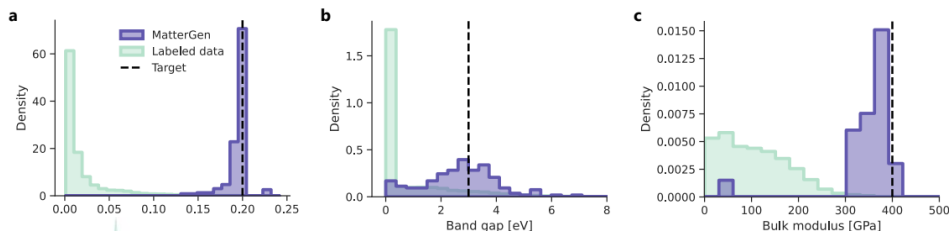
背景介绍

MatterGen 微软的谢天团队 2023

CDVAE的后续工作

For each property in Fig. 5(a-c), we generate 512 samples with our fine-tuned model by conditioning on a value of $0.2 \text{ \AA}^{-3}$ for magnetic density, 3.0 eV for band gap, and 400 GPa for bulk modulus. We relax those structures using the MLFF and filter the relaxed structures by stability and uniqueness. We then relax the remaining structures with DFT and filter by stability, uniqueness, and novelty. Finally, we compute the desired property of the remaining structures using DFT, and filter out structures where we consider the computed property to be an outlier, i.e., high bulk modulus values with more than 600 GPa and magnetic density values with more than $0.3 \text{ \AA}^{-3}$. For more details on the DFT calculations, see Appendix C.2. After all filters and DFT computations, we obtain 251 S.U.N. structures for magnetic density, 142 for band gap, and 22 for bulk modulus.

Pre-training with structure data Fine-tuning with labeled data for condition c



SLICES
晶体编码

SLICES
晶体解码

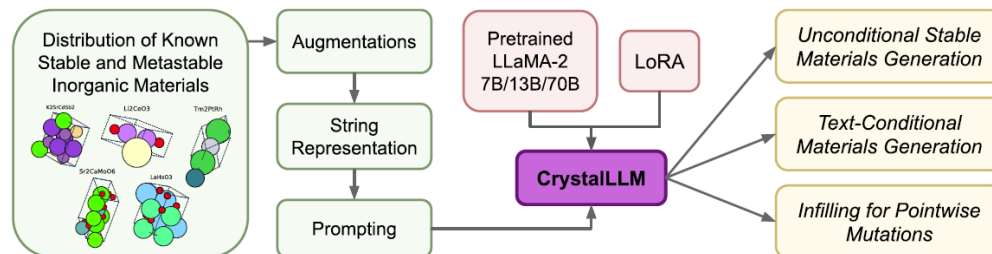
MatterGPT

不开源

架构复杂，不容易理解

结果经过筛选

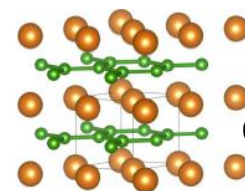
CrystaLLM 2024



CIF文本+ 提示词

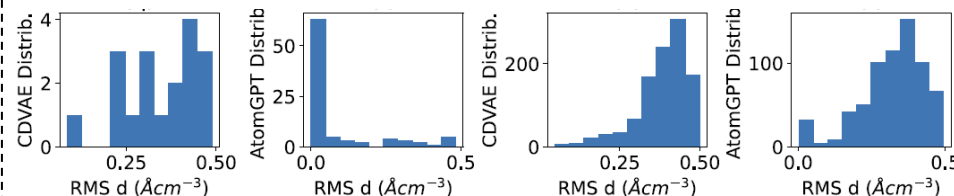
CIF不具备不变性，学习效率低

AtomGPT 2024



Text
descriptions

Alpaca input:
{Instruction: Below is a description of a superconductor material.,
Input: The chemical formula is MgB_2 . The T_c supercon is 32.685. Generate atomic structure description with lattice lengths, angles, coordinates and atom types.,
Output: 3.07 3.07 3.51\n90 90 120\nMg 0.000 0.000 0.000\nB 0.667 0.333 0.500\nB 0.333 0.667 0.500}

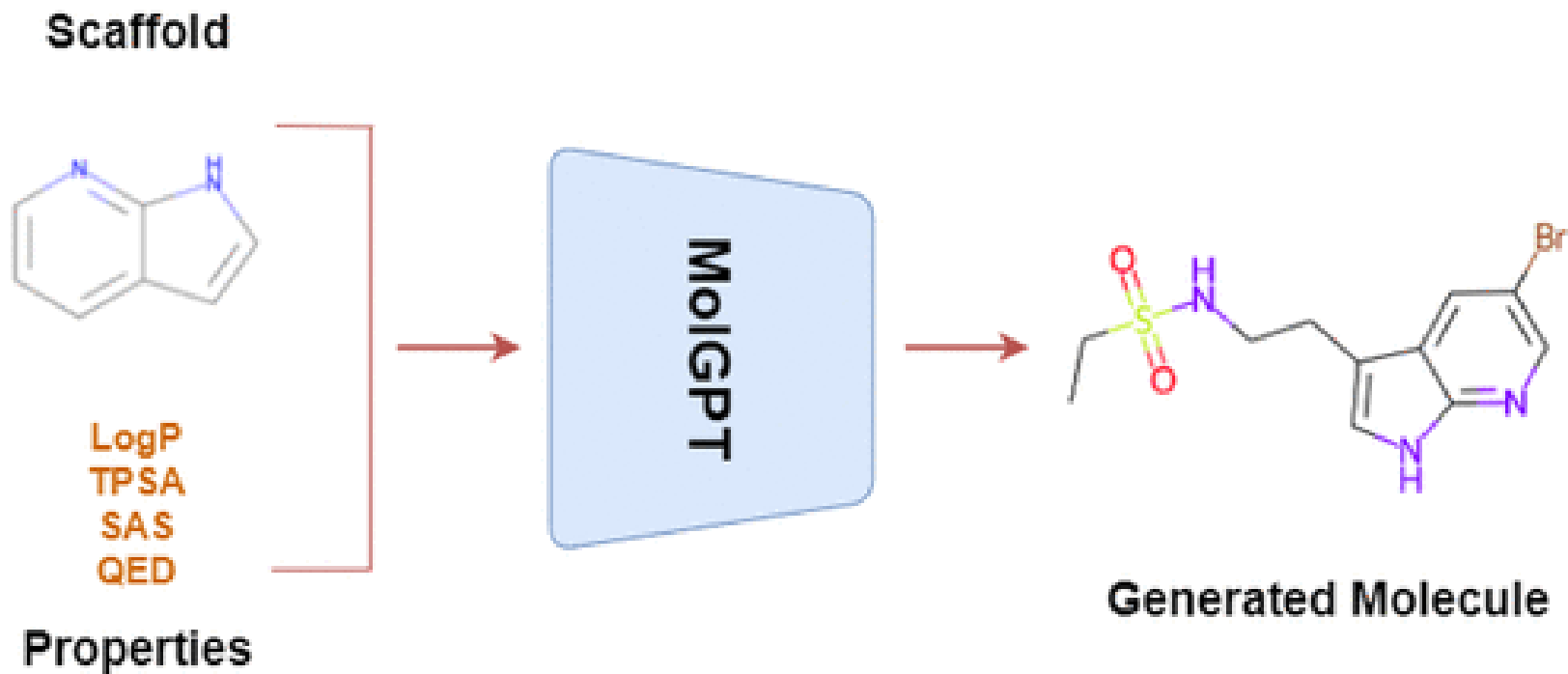


逆向设计效率低，性质控制不准确

• 来自MolGPT的启发

背景介绍

■ GPT-2 + SMILES



Bagal, Viraj, et al. "MolGPT: molecular generation using a transformer-decoder model." *Journal of Chemical Information and Modeling* 62.9 (2021): 2064-2076.

MatterGPT

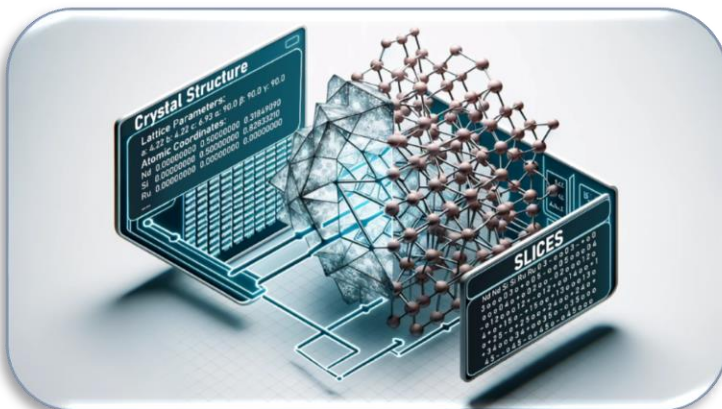
• MatterGPT的架构: GPT-2 + SLICES

背景介绍



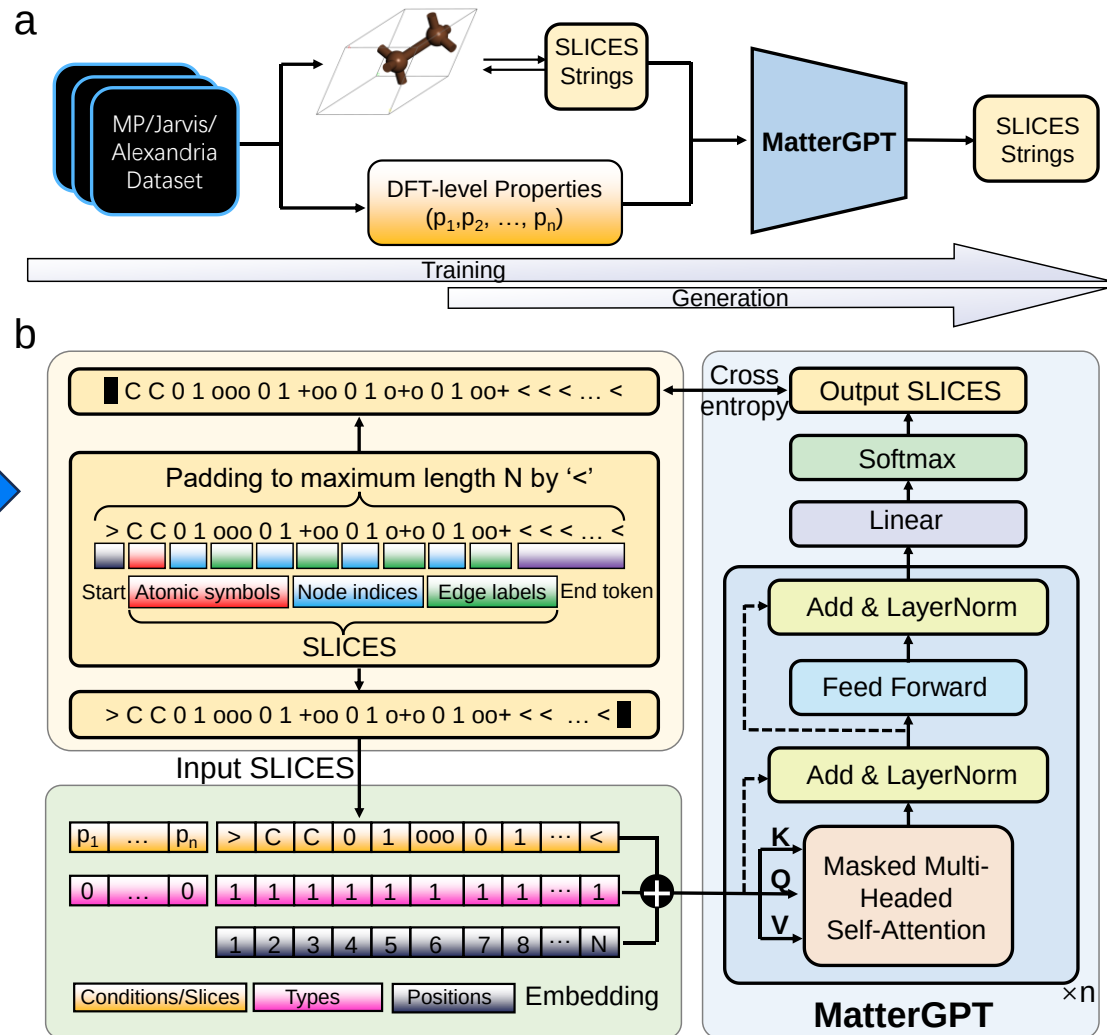
SLICES
晶体编码

SLICES
晶体解码



Crystal Language: **SLICES**

MatterGPT



• MatterGPT的模型参数

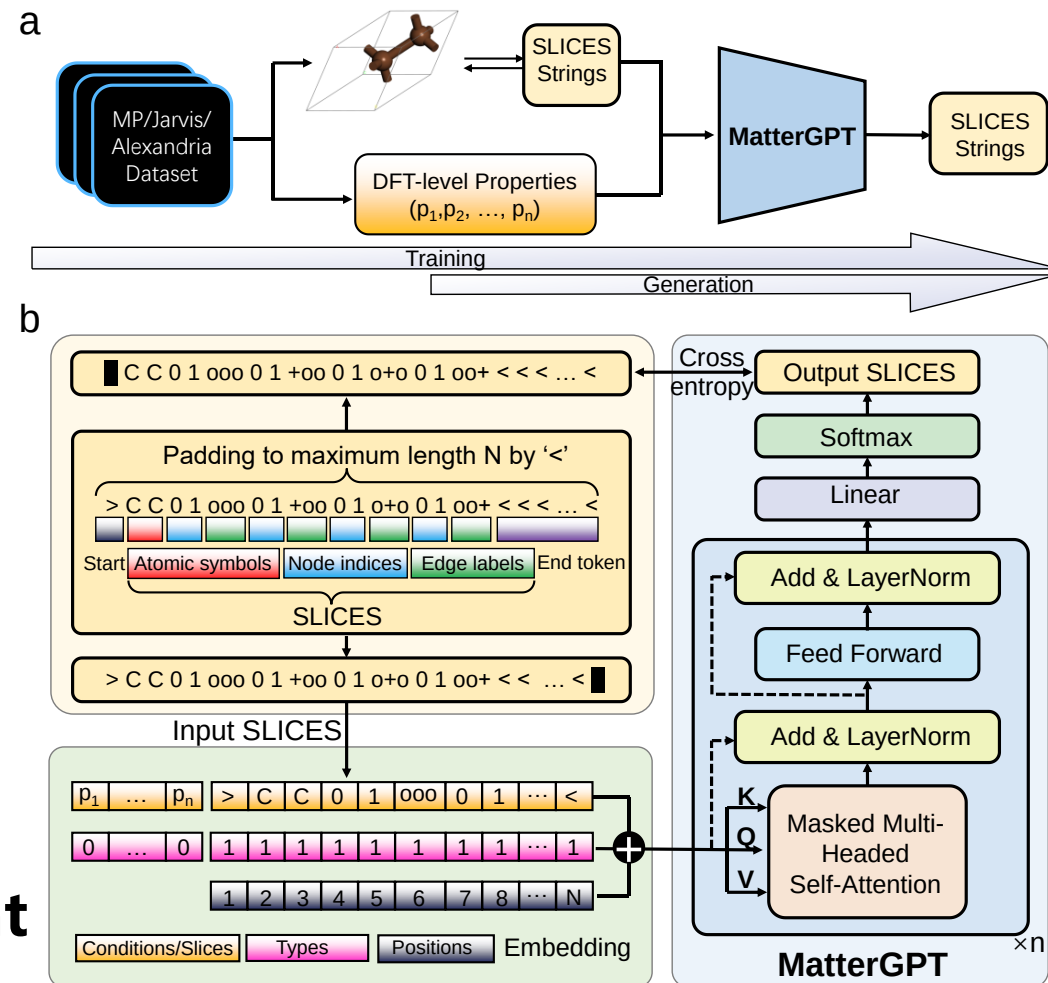
背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT

- Decoder-only model
- 嵌入维度 (n_embd) = 768
- 解码器堆叠层数(n_layer) = 12
- 多头注意力头数(n_head) = 12
- 可训练参数 $\approx$ 80M
- 分词器: 空格分词
- 位置嵌入: 可学习型 (learnable)
- 性质嵌入 : 线性层
- 性质嵌入与文本嵌入结合: Concat
- 训练任务类型: Next token pred



• 单性质逆向设计性材料-形成能

背景介绍

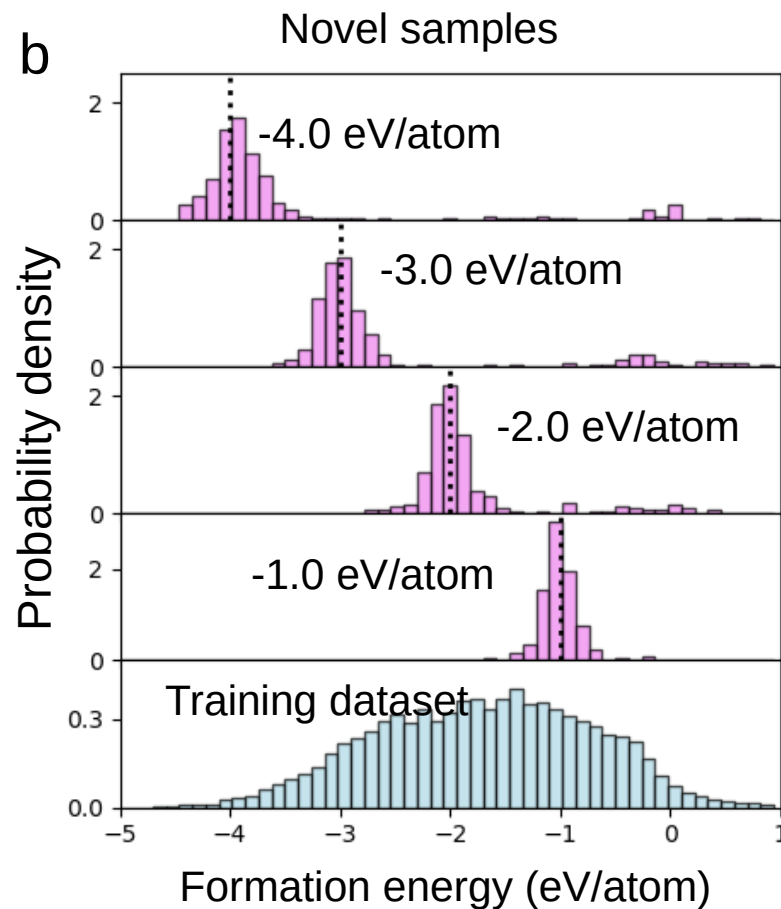
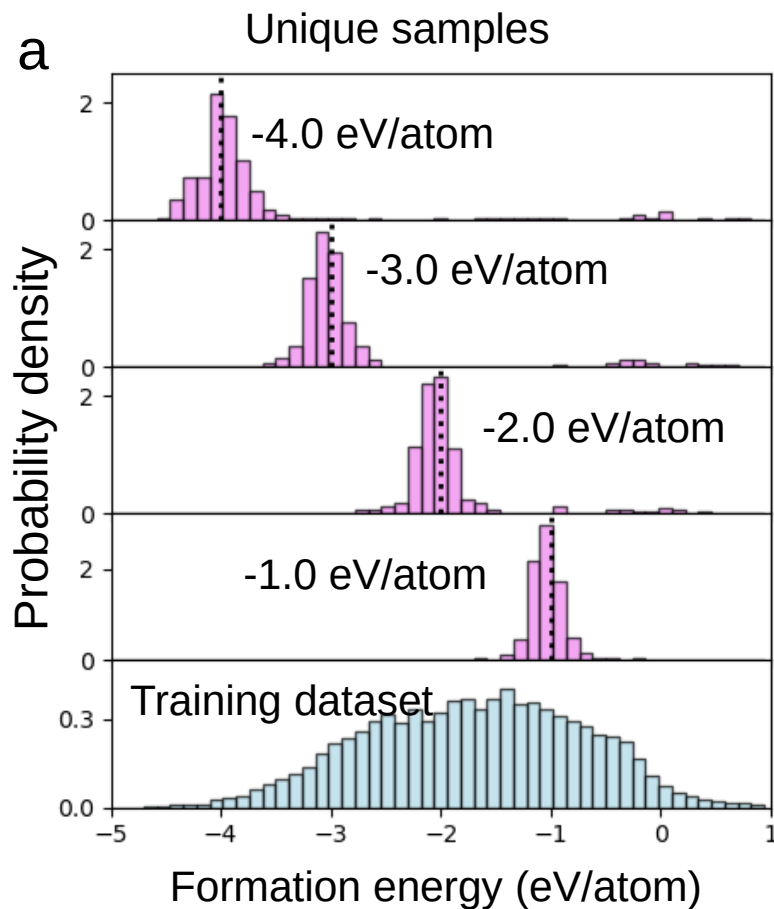
■ 训练集为28万个 [SLICES, E_{form}]

■ 指定形成能目标值为 -1.0, -2.0, -3.0, -4.0 eV/atom

SLICES
晶体编码

SLICES
晶体解码

MatterGPT



• 单性质逆向设计性材料-带隙

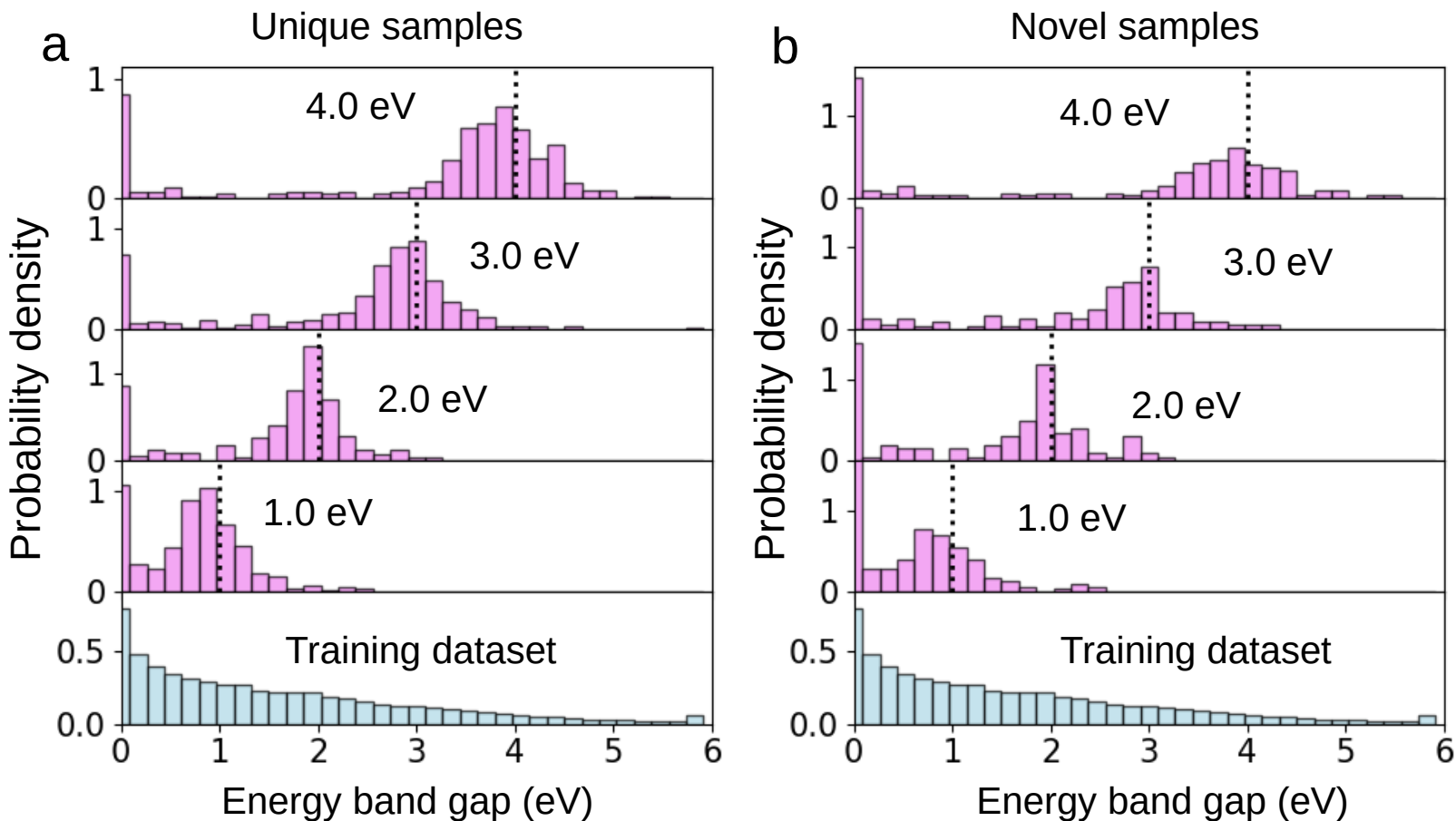
背景介绍

- 训练集为28万个 [SLICES, bandgap]
- 指定带隙目标值为 1.0, 2.0, 3.0, 4.0 eV

SLICES
晶体编码

SLICES
晶体解码

MatterGPT



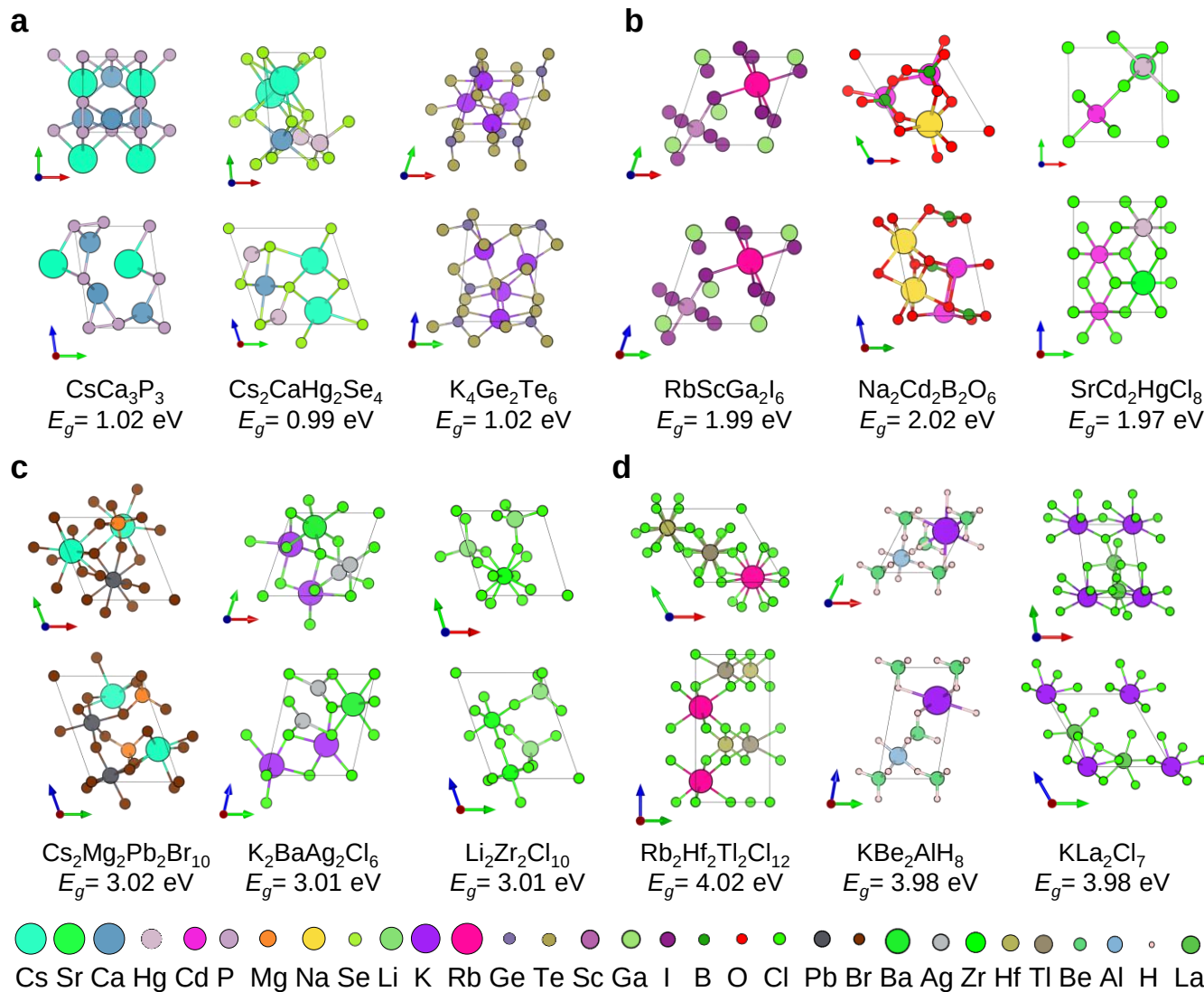
指定带隙生成的新材料结构

背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT



■ 生成的具有指定带隙的新材料结构图

■ a-d: $E_g = 1.0, 2.0, 3.0, 4.0$ eV

指定带隙生成的新材料能带图

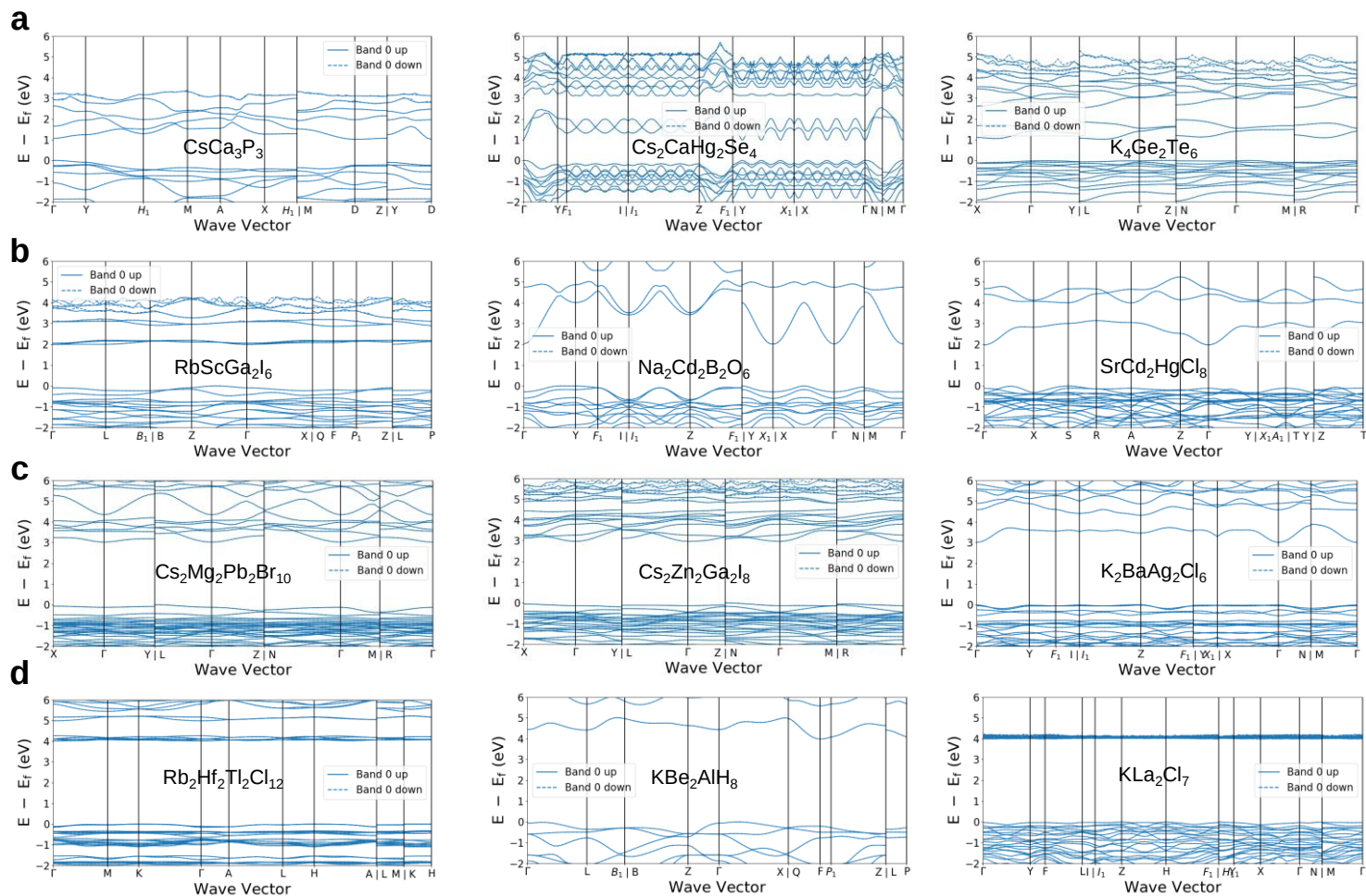
背景介绍

SLICES
晶体编码

SLICES
晶体解码

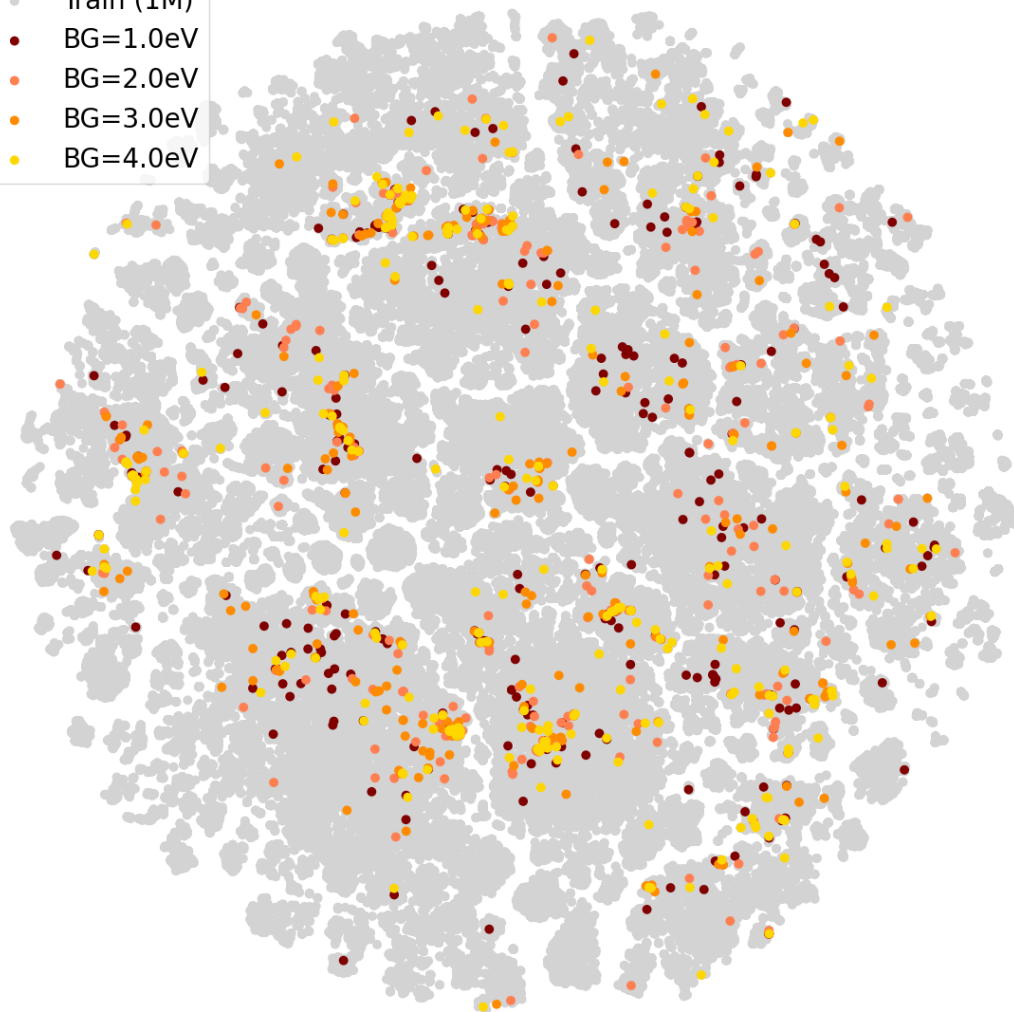
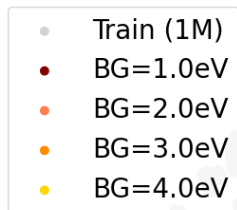
MatterGPT

■ PBE 能带图 a-d: $E_g = 1.0, 2.0, 3.0, 4.0$ eV



指定带隙生成的新材料结构的t-SNE图

背景介绍



SLICES
晶体编码

SLICES
晶体解码

MatterGPT

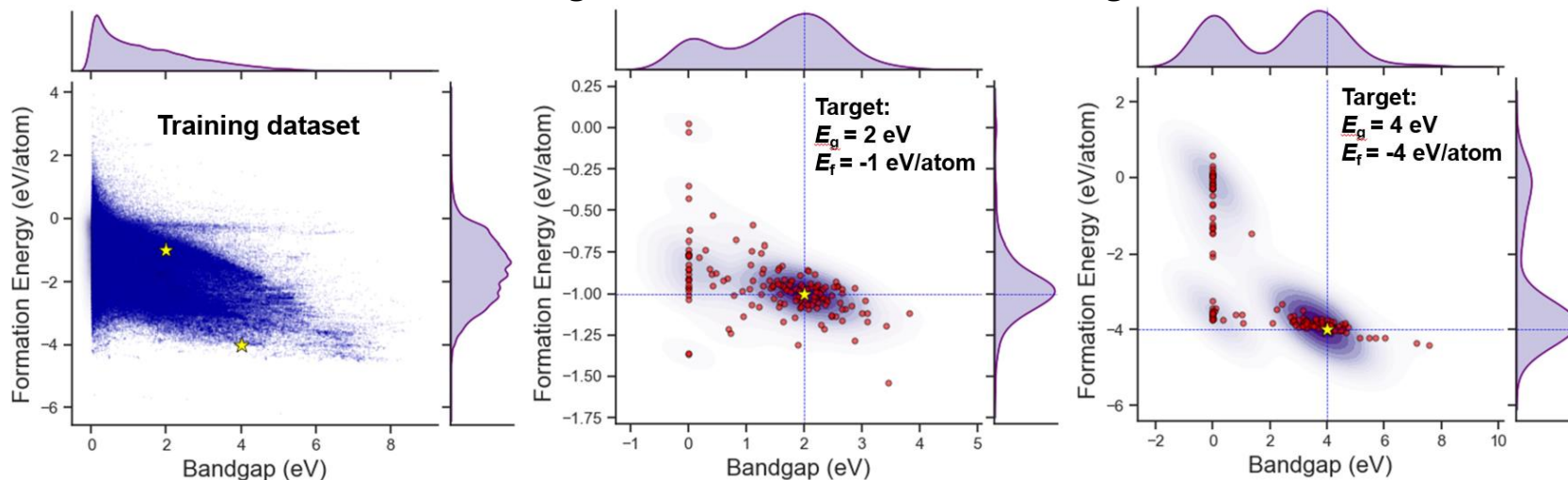
- 不同带隙目标(1.0-4.0eV)下生成的晶体分布在整个空间中。
- 生成多样化的结构
- 具有良好的**泛化能力**

多性质逆向设计新材料

背景介绍

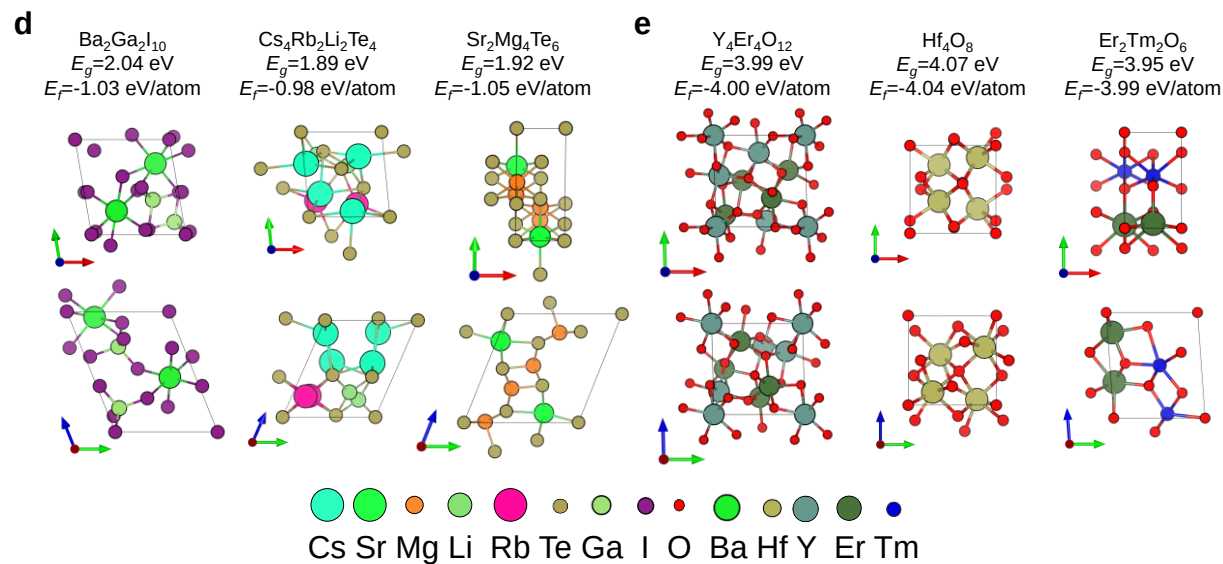
■ 训练集为28万个[SLICES, E_g , E_f]

■ 指定目标(E_g , E_f)



SLICES
晶体编码

SLICES
晶体解码



MatterGPT

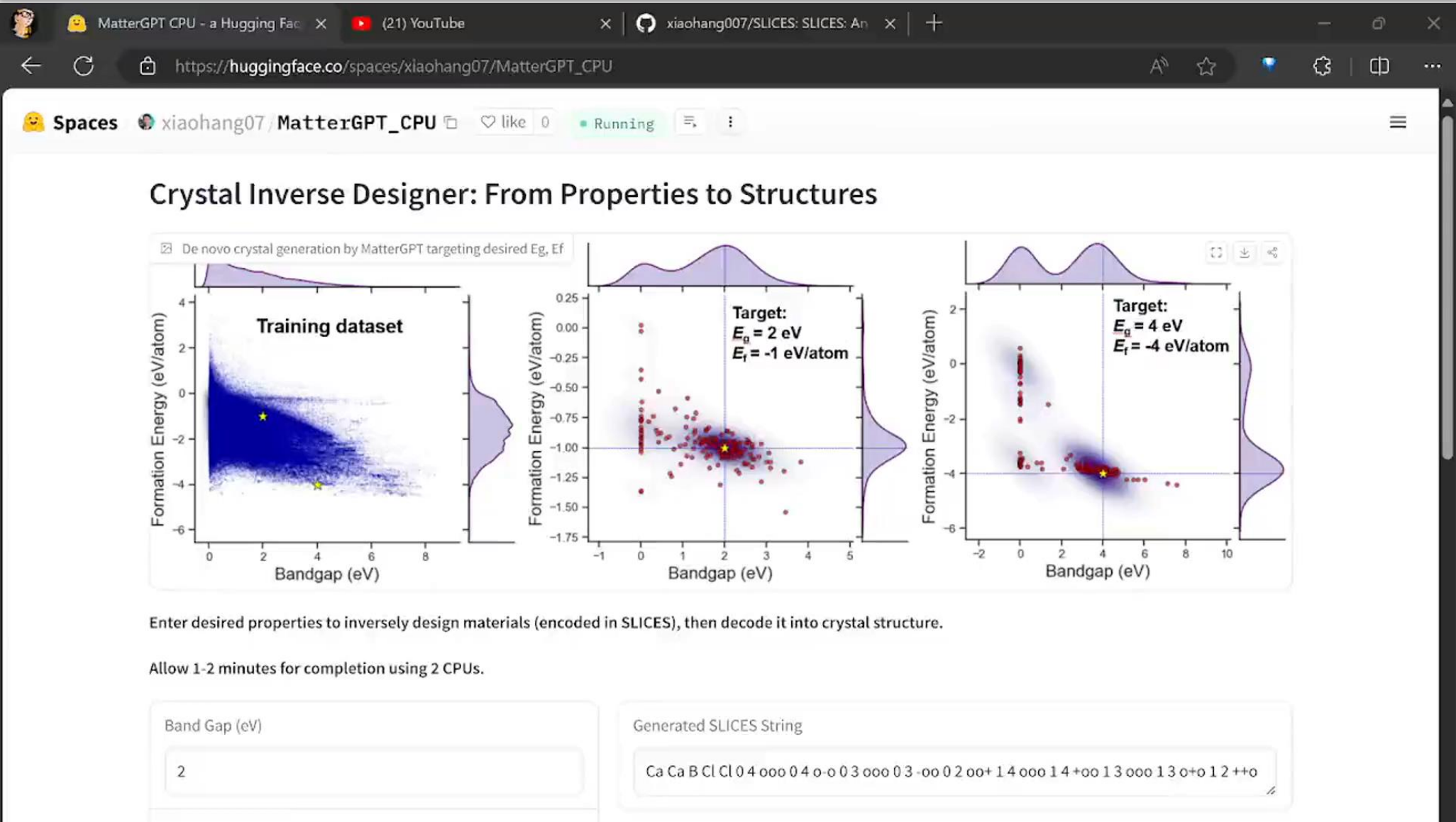
• Huggingface Space 在线演示

背景介绍

SLICES
晶体编码

SLICES
晶体解码

MatterGPT



• MatterGPT总结

背景介绍

SLICES
晶体编码

SLICES
晶体解码

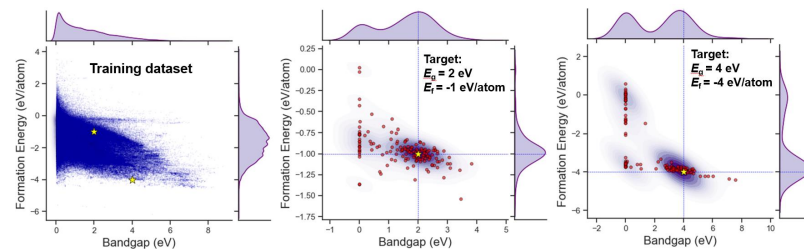
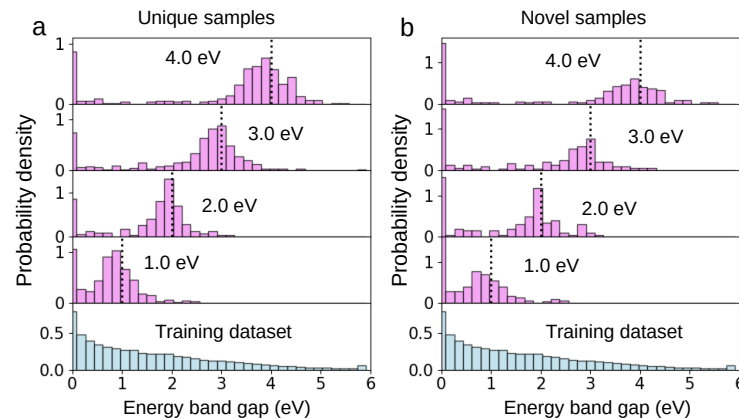
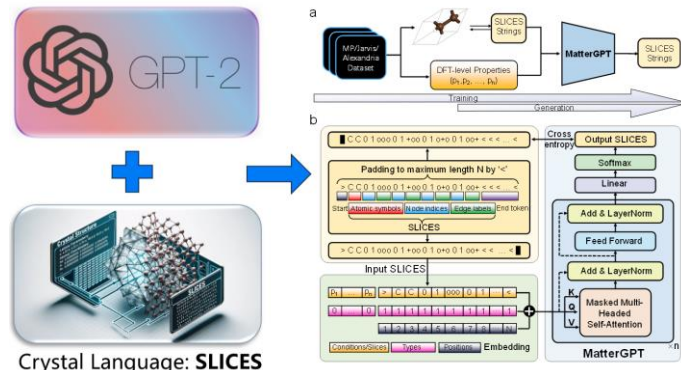
MatterGPT

- 支持多性质逆向设计（目标->新材料）
- 对生成的新材料的性质控制能力好
- 架构简单，tokenizer简单，容易学习
- 开源：训练集+训练代码+模型+参数
- 学习具有不变性的SLICES语言
- 模型参数规模小：80M（硬盘占用300M）

晶格敏感性质的逆向设计能力有提高的空间

未找到合适的预训练+微调方案

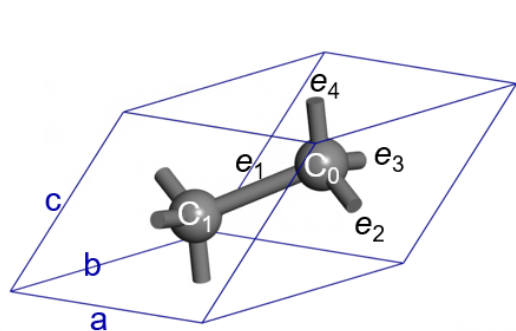
暂不支持低维材料



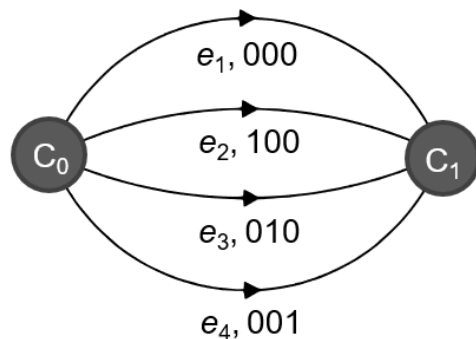
• 后续工作-扩展SLICES以提高可逆率

背景介绍

- 引入晶格常数 (95.97% - > 97.76%)、Wyckoff 信息等具有不变性的信息



晶体



晶体图

SLICES 字符串

e ₁		e ₂		e ₃		e ₄								
C	C	0	1	0	0	0	1	0	0	0	1	0	0	+
元素符号		原子序号		化学键偏移向量										

SLICES 字符串

SLICES
晶体解码

SLICES+ 字符串

						e ₁		e ₂		e ₃		e ₄								
2.51	2.51	2.51	60.0	60.0	60.0	C	C	0	1	0	0	0	1	0	0	0	1	0	0	+
晶 格 常 数						元素符号		原子序号		化学键偏移向量										

SLICES+ 字符串

MatterGPT

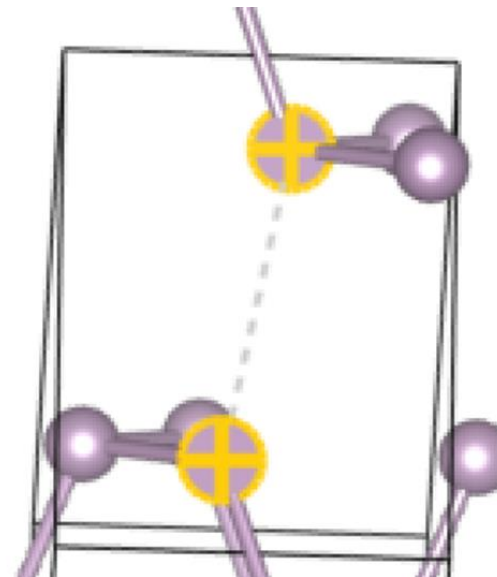
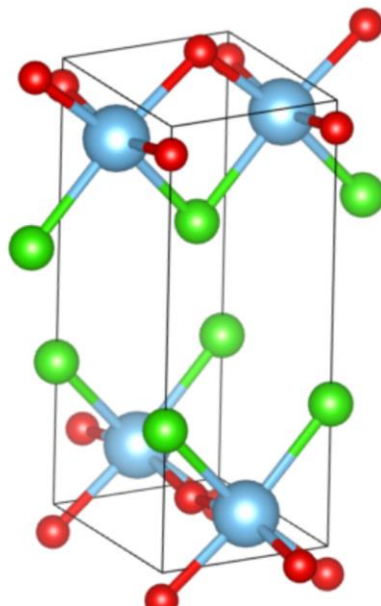
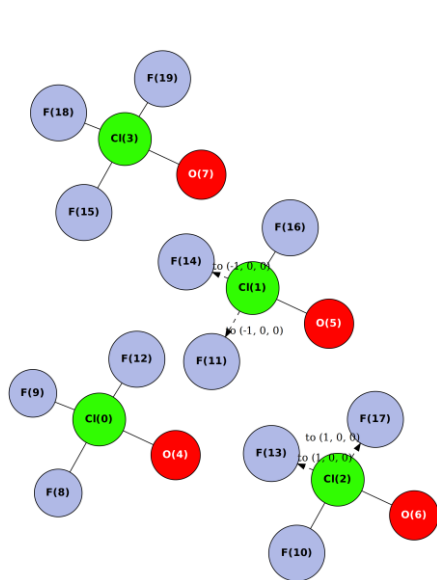
• 后续工作-改进SLICES解码算法使其适用于低维材料

背景介绍

- 引入vdW虚拟键，使得低维材料的晶体图变为连通图且具有三个方向的偏移矢量

SLICES
晶体编码

SLICES
晶体解码



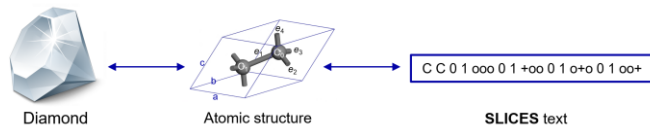
3.27 3.97 8.21 90.0 90.0 90.0 Ti Ti Cl Cl O O O 4 oo- 0 4 +o- 0 3 ooo 0 3 +oo 0 5
o-o 0 5 ooo 1 2 -oo 1 2 ooo 1 5 -o+ 1 5 oo+ 1 4 ooo 1 4 o+o vdw 2 3 ooo 2 3 o+o
2 3 +oo 2 3 ++o

MatterGPT

- 对MP20中的2000个二维材料，实现75.2%的解码成功率

语言：材料编码

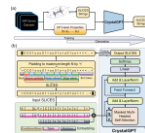
■ SLICES



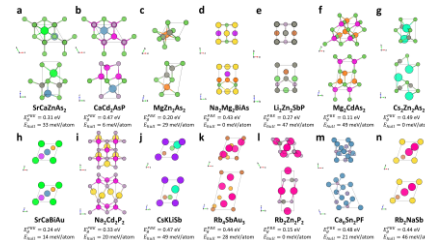
平台：晶体语言模型

■ MatterGPT

开源，架构简单

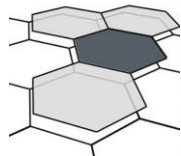


应用：新型功能材料

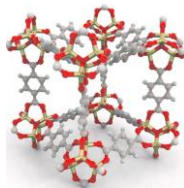


抛砖引玉，共同开发

■ 2DSLICES



■ MOFSLICES



■ MatterLlama



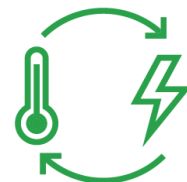
■ MatterQwen



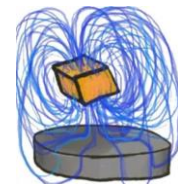
■ SLICES-GNN



■ 热电材料



■ 超导材料





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University 香港 Hong Kong

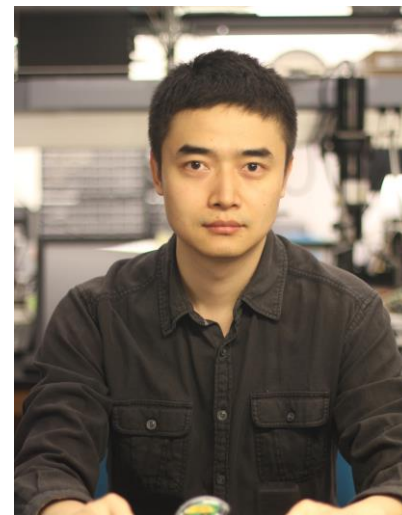
致 谢



陈曦 教授
岭南大学
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陈炎 助理教授
西安交通大学
航天航空学院



王雷 教授
南京大学
物理学院



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恳请各位专家批评指正！
谢谢！

Liberal Arts in the Digital Age



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SLICES新材料挖掘机

香港岭南大学 跨学科学院 肖航

